多元复合函数的求导法则

一元复合函数　$y = f(u),\ u = \varphi(x)$

求导法则　$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y}{\mathrm{d}u} \cdot \dfrac{\mathrm{d}u}{\mathrm{d}x}$

微分法则　$\mathrm{d}y = f'(u)\,\mathrm{d}u = f'(u)\varphi'(x)\,\mathrm{d}x$

本节内容：

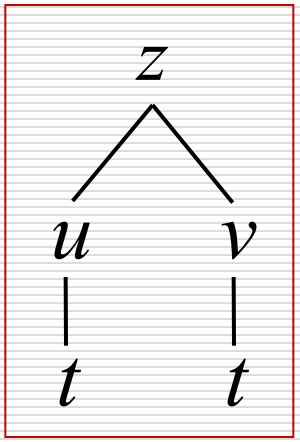
一、多元复合函数求导的链式法则

二、多元复合函数的全微分

一、多元复合函数求导的链式法则

定理. 若函数 $u=\varphi(t), v=\psi(t)$ 在点 t 可导, $z=f(u,v)$ 在点 (u,v) 处偏导连续, 则复合函数 $z=f(\varphi(t),\psi(t))$ 在点 t 可导, 且有链式法则

$$\frac{\mathrm{d}z}{\mathrm{d}t}=\frac{\partial z}{\partial u}\cdot\frac{\mathrm{d}u}{\mathrm{d}t}+\frac{\partial z}{\partial v}\cdot\frac{\mathrm{d}v}{\mathrm{d}t}$$



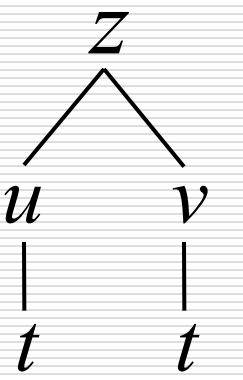
证: 设 t 取增量 $\triangle t$, 则相应中间变量有增量 $\triangle u, \triangle v$,

$$\Delta z=\frac{\partial z}{\partial u}\Delta u+\frac{\partial z}{\partial v}\Delta v+o(\rho)\quad\left(\rho=\sqrt{(\Delta u)^2+(\Delta v)^2}\right)$$

$$\frac{\Delta z}{\Delta t}=\frac{\partial z}{\partial u}\frac{\Delta u}{\Delta t}+\frac{\partial z}{\partial v}\frac{\Delta v}{\Delta t}+\frac{o(\rho)}{\Delta t}\quad(\rho=\sqrt{(\Delta u)^2+(\Delta v)^2})$$

令 $\Delta t\to 0$, 则有 $\Delta u\to 0,\ \Delta v\to 0,$

$$\frac{\Delta u}{\Delta t}\to\frac{\mathrm{d}u}{\mathrm{d}t},\qquad \frac{\Delta v}{\Delta t}\to\frac{\mathrm{d}v}{\mathrm{d}t}$$



$$\frac{o(\rho)}{\Delta t}=\frac{o(\rho)}{\rho}\sqrt{(\frac{\Delta u}{\Delta t})^2+(\frac{\Delta v}{\Delta t})^2}\to 0$$

(△t＜0 时, 根式前加“－”号)

$$\frac{\mathrm{d}z}{\mathrm{d}t}=\frac{\partial z}{\partial u}\cdot\frac{\mathrm{d}u}{\mathrm{d}t}+\frac{\partial z}{\partial v}\cdot\frac{\mathrm{d}v}{\mathrm{d}t}$$

(全导数公式)

说明： 若定理中 $f(u,v)$ 在点 (u,v) **偏导数连续**减弱为**偏导数存在**，则定理结论不一定成立.

例如： $z=f(u,v)=\begin{cases}\dfrac{u^2v}{u^2+v^2}, & u^2+v^2\neq 0\\ 0, & u^2+v^2=0\end{cases}$

$u=t,\quad v=t$

易知: $\left.\dfrac{\partial z}{\partial u}\right|_{(0,0)}=f_u(0,0)=0,\ \left.\dfrac{\partial z}{\partial v}\right|_{(0,0)}=f_v(0,0)=0$

但复合函数 $z=f(t,t)=\dfrac{t}{2}$

$$\frac{\mathrm{d}z}{\mathrm{d}t}=\frac{1}{2}\neq\frac{\partial z}{\partial u}\cdot\frac{\mathrm{d}u}{\mathrm{d}t}+\frac{\partial z}{\partial v}\cdot\frac{\mathrm{d}v}{\mathrm{d}t}=0\cdot 1+0\cdot 1=0$$

一、多元复合函数求导的链式法则

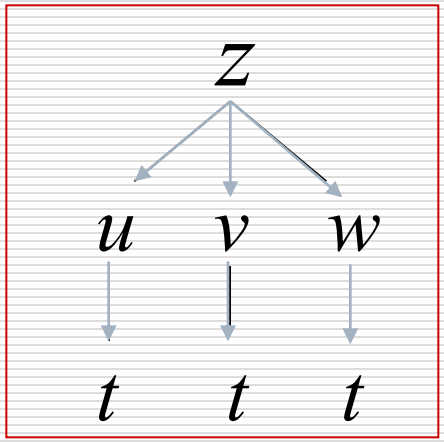
定理推广： 设下面所涉及的函数都可微 .

1) 中间变量多于两个的情形. 例如, $z=f(u,v,w)$,

$$u=\varphi(t),\ v=\psi(t),\ w=\omega(t)$$

$$\frac{\mathrm{d}z}{\mathrm{d}t}=\frac{\partial z}{\partial u}\cdot\frac{\mathrm{d}u}{\mathrm{d}t}+\frac{\partial z}{\partial v}\cdot\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{\partial z}{\partial w}\cdot\frac{\mathrm{d}w}{\mathrm{d}t}$$

$$=f_1'\varphi'+f_2'\psi'+f_3'\omega'$$

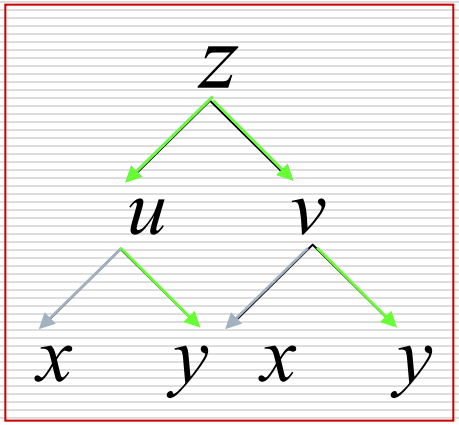


一、多元复合函数求导的链式法则

定理推广： 设下面所涉及的函数都可微 .

2) 中间变量是多元函数的情形.例如,

$$z=f(u,v),\quad u=\varphi(x,y),\quad v=\psi(x,y)$$



$$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}=f_1'\varphi_1'+f_2'\psi_1'$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}=f_1'\varphi_2'+f_2'\psi_2'$$

口诀： 分段用乘, 分叉用加, 单路全导, 叉路偏导

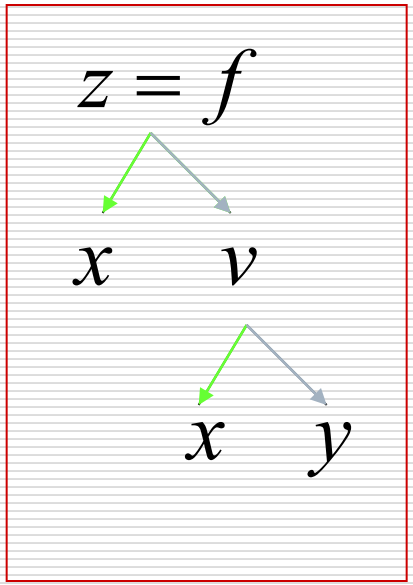
特例

又如， $z=f(x,v),\ v=\psi(x,y)$

当它们都具有可微条件时, 有

$$\frac{\partial z}{\partial x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=f_1'+f_2'\psi_1'$$

$$\frac{\partial z}{\partial y}=\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=f_2'\psi_2'$$



注意： 这里 $\frac{\partial z}{\partial x}$ 与 $\frac{\partial f}{\partial x}$ 不同， $\frac{\partial z}{\partial x}$ 表示固定 y 对 x 求导，

$\frac{\partial f}{\partial x}$ 表示固定 v 对 x 求导

例题

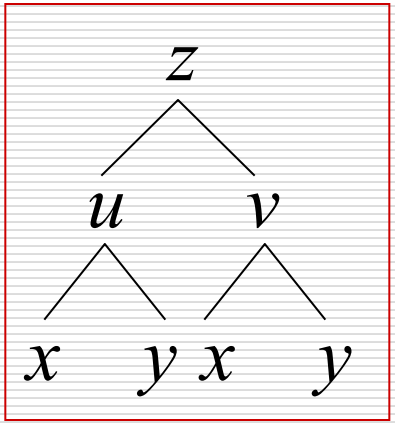
例1. 设 $z = e^u \sin v,\ u = xy,\ v = x + y$, 求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

解: $$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$= e^u \sin v \cdot y + e^u \cos v \cdot 1$$

$$= e^{xy}[y\cdot\sin(x+y) + \cos(x+y)]$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$



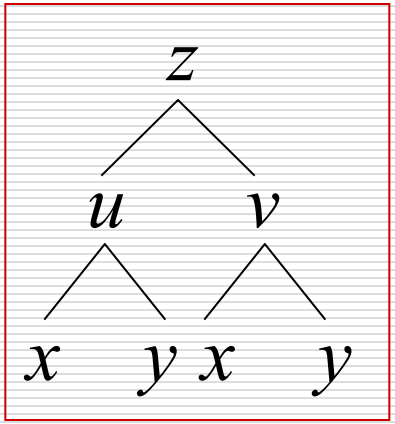
例题

例1. 设 $z = e^u \sin v,\ u = xy,\ v = x + y$, 求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

续解: $\dfrac{\partial z}{\partial y} = \dfrac{\partial z}{\partial u} \cdot \dfrac{\partial u}{\partial y} + \dfrac{\partial z}{\partial v} \cdot \dfrac{\partial v}{\partial y}$

$= e^u \sin v \cdot x + e^u \cos v \cdot 1$

$= e^{xy}[x \cdot \sin(x + y) + \cos(x + y)]$



例题

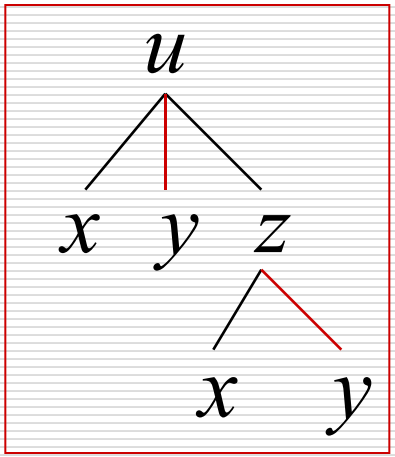
例2. $u = f(x, y, z) = e^{x^2+y^2+z^2}$, $z = x^2 \sin y$, 求 $\dfrac{\partial u}{\partial x}, \dfrac{\partial u}{\partial y}$

解: $\dfrac{\partial u}{\partial x} = \dfrac{\partial f}{\partial x} + \dfrac{\partial f}{\partial z} \cdot \dfrac{\partial z}{\partial x}$

$= 2xe^{x^2+y^2+z^2} + 2ze^{x^2+y^2+z^2} \cdot 2x \sin y$

$= 2x(1 + 2x^2 \sin^2 y) e^{x^2+y^2+x^4 \sin^2 y}$

$\dfrac{\partial u}{\partial y} = \dfrac{\partial f}{\partial y} + \dfrac{\partial f}{\partial z} \cdot \dfrac{\partial z}{\partial y}$



例题

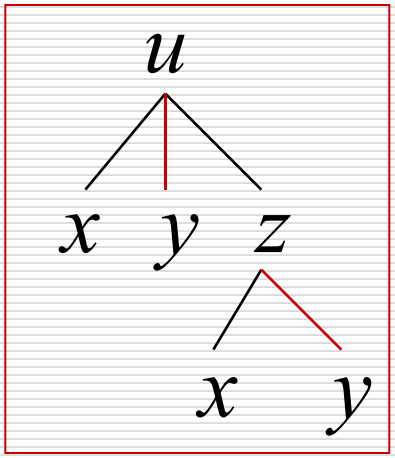
例2. $u = f(x, y, z) = e^{x^2+y^2+z^2}$, $z = x^2 \sin y$, 求 $\dfrac{\partial u}{\partial x}, \dfrac{\partial u}{\partial y}$

续解:

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y}$$

$$= 2ye^{x^2+y^2+z^2} + 2ze^{x^2+y^2+z^2} \cdot x^2 \cos y$$

$$= 2\left(y + x^4 \sin y \cos y\right) e^{x^2+y^2+x^4 \sin^2 y}$$



例题

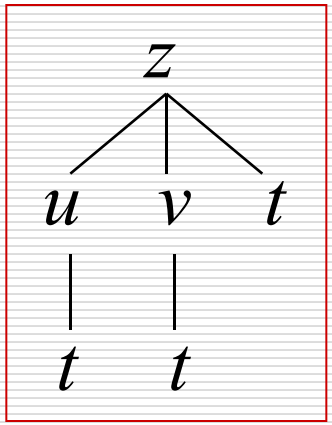
例3. 设 $z=uv+\sin t$, $u=e^t$, $v=\cos t$, 求全导数 $\dfrac{\mathrm{d}z}{\mathrm{d}t}$.

解:

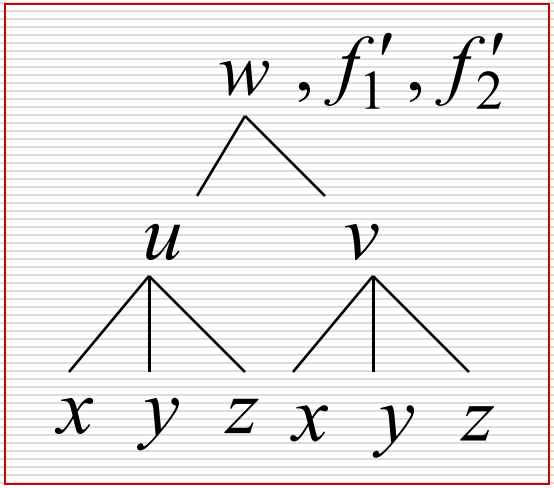
$$\frac{\mathrm{d}z}{\mathrm{d}t}=\frac{\partial z}{\partial u}\cdot\frac{\mathrm{d}u}{\mathrm{d}t}+\frac{\partial z}{\partial v}\cdot\frac{\mathrm{d}v}{\mathrm{d}t}+\frac{\partial z}{\partial t}$$

$$=v\,e^t-u\sin t+\cos t$$

$$=e^t(\cos t-\sin t)+\cos t$$



例4. 设 $w = f(x+y+z, xyz)$, f 具有二阶连续偏导数, 求 $\dfrac{\partial w}{\partial x}, \dfrac{\partial^2 w}{\partial x \partial z}$.



解: 令 $u = x+y+z$, $v = xyz$, 则

$$w = f(u, v)$$

$$\frac{\partial w}{\partial x} = f_1' \cdot 1 + f_2' \cdot yz$$

$$= f_1'(x+y+z, xyz) + yz\, f_2'(x+y+z, xyz)$$

$$\frac{\partial^2 w}{\partial x \partial z} = f_{11}'' \cdot 1 + f_{12}'' \cdot xy + y\, f_2' + yz[\, f_{21}'' \cdot 1 + f_{22}'' \cdot xy]$$

$$= f_{11}'' + y(x+z) f_{12}'' + xy^2 z\, f_{22}'' + y\, f_2'$$

练习

1.求下列函数的一阶偏导数：

$$u = f\left(\frac{x}{y}, \frac{y}{z}\right).$$

2.求下列函数的二阶偏导数：

$$z = f(u, x, y),\ u = xe^{y}.$$

练习-答案

1. $u = f\left(\frac{x}{y}, \frac{y}{z}\right).$

解：
$$\frac{\partial u}{\partial x} = f_1' \cdot \frac{1}{y} = \frac{1}{y} f_1'$$

$$\frac{\partial u}{\partial y} = f_1' \cdot \left(-\frac{x}{y^2}\right) + f_2' \cdot \frac{1}{z} = -\frac{x}{y^2} f_1' + \frac{1}{z} f_2'$$

$$\frac{\partial u}{\partial z} = f_2' \cdot \left(-\frac{y}{z^2}\right) = -\frac{y}{z^2} f_2'$$

练习-答案

$$z = f(u, x, y),\ u = xe^{y}$$

解：
$$\frac{\partial z}{\partial x} = f_1' \cdot e^{y} + f_2'$$

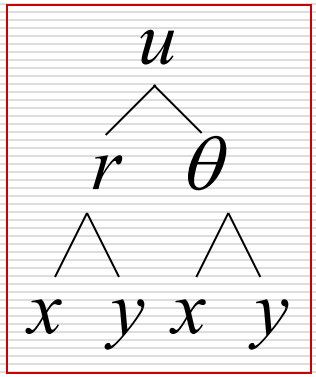
$$\frac{\partial^2 z}{\partial x \partial y} = e^{y} \cdot f_1' + x e^{2y} \cdot f_{11}'' + e^{y} \cdot f_{13}'' + x e^{y} \cdot f_{21}'' + f_{23}''$$

其余解略.

例5. 设 $u=f(x,y)$ 二阶偏导数连续, 求下列表达式在极坐标系下的形式 (1) $(\frac{\partial u}{\partial x})^2+(\frac{\partial u}{\partial y})^2$, (2) $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}$

解: 已知 $x=r\cos\theta$, $y=r\sin\theta$, 则

$$r=\sqrt{x^2+y^2},\ \theta=\arctan\frac{y}{x}$$



(1) $$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial x}+\frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial x}$$

$$\frac{\partial r}{\partial x}=\frac{x}{r},\quad \frac{\partial \theta}{\partial x}=\frac{\frac{-y}{x^2}}{1+(\frac{y}{x})^2}=\frac{-y}{x^2+y^2}$$

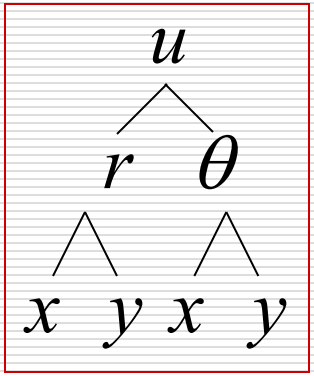
$$=\frac{\partial u}{\partial r}\frac{x}{r}-\frac{\partial u}{\partial \theta}\frac{y}{r^2}=\frac{\partial u}{\partial r}\cos\theta-\frac{\partial u}{\partial \theta}\frac{\sin\theta}{r}$$

$$\frac{\partial u}{\partial y}=\frac{\partial u}{\partial r}\frac{\partial r}{\partial y}+\frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial y}$$

$$\frac{\partial r}{\partial y}=\frac{y}{r},\quad \frac{\partial \theta}{\partial y}=\frac{\frac{1}{x}}{1+(\frac{y}{x})^2}=\frac{x}{x^2+y^2}$$

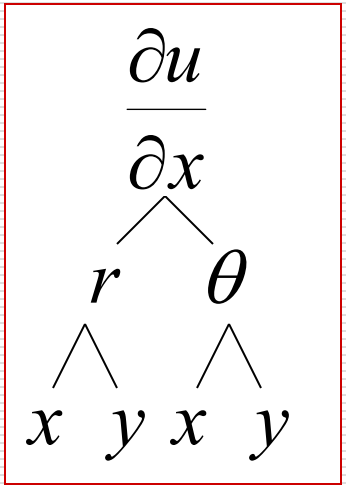
$$=\frac{\partial u}{\partial r}\frac{y}{r}+\frac{\partial u}{\partial \theta}\frac{x}{r^2}$$

$$=\frac{\partial u}{\partial r}\sin\theta+\frac{\partial u}{\partial \theta}\frac{\cos\theta}{r}$$



$$\therefore\quad (\frac{\partial u}{\partial x})^2+(\frac{\partial u}{\partial y})^2=(\frac{\partial u}{\partial r})^2+\frac{1}{r^2}(\frac{\partial u}{\partial \theta})^2$$

$$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial r}\cos\theta-\frac{\partial u}{\partial \theta}\frac{\sin\theta}{r}$$



注意利用已有公式

(2) $$\frac{\partial^2 u}{\partial x^2}=\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)\cdot\cos\theta-\frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial x}\right)\frac{\sin\theta}{r}$$

$$=\frac{\partial}{\partial r}\left(\frac{\partial u}{\partial r}\cos\theta-\frac{\partial u}{\partial \theta}\frac{\sin\theta}{r}\right)\cdot\cos\theta$$

$$-\frac{\partial}{\partial \theta}\left(\frac{\partial u}{\partial r}\cos\theta-\frac{\partial u}{\partial \theta}\frac{\sin\theta}{r}\right)\cdot\frac{\sin\theta}{r}$$

$$=\frac{\partial^2 u}{\partial r^2}\cos^2\theta-2\frac{\partial^2 u}{\partial r\partial \theta}\frac{\sin\theta\cos\theta}{r}+\frac{\partial^2 u}{\partial \theta^2}\frac{\sin^2\theta}{r^2}$$

$$+\frac{\partial u}{\partial \theta}\frac{2\sin\theta\cos\theta}{r^2}+\frac{\partial u}{\partial r}\frac{\sin^2\theta}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial r^2}\cos^2\theta - 2\frac{\partial^2 u}{\partial r \partial \theta}\frac{\sin\theta\cos\theta}{r} + \frac{\partial^2 u}{\partial \theta^2}\frac{\sin^2\theta}{r^2}$$

$$+\frac{\partial u}{\partial \theta}\frac{2\sin\theta\cos\theta}{r^2} + \frac{\partial u}{\partial r}\frac{\sin^2\theta}{r}$$

同理可得

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2}\sin^2\theta + 2\frac{\partial^2 u}{\partial r \partial \theta}\frac{\sin\theta\cos\theta}{r} + \frac{\partial^2 u}{\partial \theta^2}\frac{\cos^2\theta}{r^2}$$

$$-\frac{\partial u}{\partial \theta}\frac{2\sin\theta\cos\theta}{r^2} + \frac{\partial u}{\partial r}\frac{\cos^2\theta}{r}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}$$

$$= \frac{1}{r^2}\left[r\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{\partial^2 u}{\partial \theta^2}\right]$$

二、多元复合函数的全微分

设函数 $z=f(u,v),u=\varphi(x,y),v=\psi(x,y)$ 都可微，则复合函数 $z=f(\varphi(x,y),\psi(x,y))$ 的全微分为

$$
\begin{aligned}
\mathrm{d}z &= \frac{\partial z}{\partial x}\mathrm{d}x+\frac{\partial z}{\partial y}\mathrm{d}y \\
&= \left(\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}\right)\mathrm{d}x+\left(\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}\right)\mathrm{d}y \\
&= \frac{\partial z}{\partial u}\left(\frac{\partial u}{\partial x}\mathrm{d}x+\frac{\partial u}{\partial y}\mathrm{d}y\right)+\frac{\partial z}{\partial v}\left(\frac{\partial v}{\partial x}\mathrm{d}x+\frac{\partial v}{\partial y}\mathrm{d}y\right) \\
&= \frac{\partial z}{\partial u}\mathrm{d}u+\frac{\partial z}{\partial v}\mathrm{d}v
\end{aligned}
$$

二、多元复合函数的全微分

$$\mathrm{d}z = \frac{\partial z}{\partial x}\mathrm{d}x + \frac{\partial z}{\partial y}\mathrm{d}y$$

$$= \frac{\partial z}{\partial u}\mathrm{d}u + \frac{\partial z}{\partial v}\mathrm{d}v$$

可见无论 u , v 是自变量还是中间变量, 其全微分表达形式都一样, 这性质叫做**全微分形式不变性.**

例题

例1. 设 $z = e^u \sin v,\ u = xy,\ v = x + y$, 求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

例 6. 利用全微分形式不变性再解例1.

解:

$$\begin{aligned}
\mathrm{d}z &= \mathrm{d}(e^u \sin v) \\
&= e^u \sin v\,\mathrm{d}u + e^u \cos v\,\mathrm{d}v \\
&= e^{xy}[\sin(x+y)\,\mathrm{d}(xy) + \cos(x+y)\,\mathrm{d}(x+y)] \\
&= e^{xy}[\sin(x+y)(y\mathrm{d}x + x\mathrm{d}y) + \cos(x+y)(\mathrm{d}x + \mathrm{d}y)] \\
&= e^{xy}[y\sin(x+y) + \cos(x+y)]\mathrm{d}x \\
&\quad + e^{xy}[x\sin(x+y) + \cos(x+y)]\mathrm{d}y
\end{aligned}$$

例1. 设 $z = e^u \sin v,\ u = xy,\ v = x + y$, 求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

例题

例 6. 利用全微分形式不变性再解例1.

续解: $\mathrm{d}z = \mathrm{d}(e^u \sin v)$

$$= e^{xy}[y\sin(x+y) + \cos(x+y)]\mathrm{d}x$$

$$+ e^{xy}[x\sin(x+y) + \cos(x+y)]\mathrm{d}y$$

所以 $\dfrac{\partial z}{\partial x} = e^{xy}[y \cdot \sin(x+y) + \cos(x+y)]$

$$\frac{\partial z}{\partial y} = e^{xy}[x \cdot \sin(x+y) + \cos(x+y)]$$

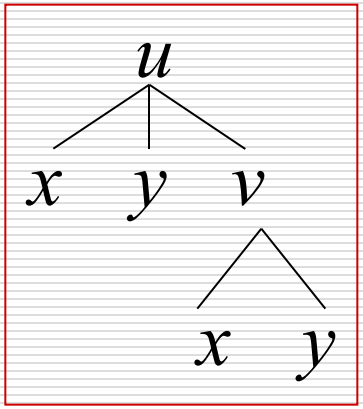
内容小结

1. 复合函数求导的链式法则

“分段用乘, 分叉用加, 单路全导, 叉路偏导”

例如, $u = f(x, y, v), v = \varphi(x, y),$

$$\frac{\partial u}{\partial x} = f_1' + f_3' \cdot \varphi_1'; \quad \frac{\partial u}{\partial y} = f_2' + f_3' \cdot \varphi_2'$$



2. 全微分形式不变性

对 $z = f(u, v)$, 不论 u, v 是自变量还是因变量,

$$\mathrm{d}z = f_u(u, v)\,\mathrm{d}u + f_v(u, v)\,\mathrm{d}v$$

作业

P82　2;　4;　6;　7;　9;　10;　12(4);